

MyCoke360 Cart Abandonment

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Overview

- Business Problem
- Topics and Findings
 - Questions 1 - 6
- Recommendations
- Q&A



Business Problem: Cart Abandonment

A significant portion of customers add products to their cart but do not complete checkout.

- Reduces online revenue.
- Suggests weakened customer loyalty.
- Suggests pain points in purchase experience.



Business Problem: Cart Abandonment

~4,000 carts were abandoned from May 2024 - May 2025.

We can potentially reduce cart abandonment by up to **70%**, leading to a profit of roughly **1 million** over a 1-year timeframe.

Let's show you how...



Cart Abandonment Flag

Process: Define customer activity segments

1. Activity between each anchor date
2. Activity from an anchor start date to a purchase
3. Activity from a purchase to the next anchor date
4. Activity from one purchase to another purchase within the same anchor date period



Cart Abandonment Flag

Process: Define abandonment and recovery flags

1. Abandoned

- a. if the “add_to_cart” event has a timestamp, but “purchase” within that segment is NaT

2. Recovered

- a. When a purchase occurs and there is a previously abandoned cart with no purchase inbetween.



Cart Abandonment Flag

purchase_date	added_to_cart	effective_end	effective_start	abandoned	purchase_segment	False_by_purchase	recovered
2024-09-11 16:52:15.890000+00:00	2024-09-11 16:49:39.743000+00:00	2024-09-11 16:52:15.890000+00:00	2024-09-03 21:55:54.690000+00:00	False	500245797.18	purchased	not applicable
2024-09-18 16:44:57.962000+00:00	2024-09-18 16:43:10.523000+00:00	2024-09-18 16:44:57.962000+00:00	2024-09-11 16:52:15.890000+00:00	False	500245797.19	purchased	not applicable
NaT	NaT	2024-09-30 00:00:00+00:00	2024-09-18 16:44:57.962000+00:00	False	500245797.20	no purchase	not applicable
NaT	2024-10-01 23:23:34.692000+00:00	2024-10-07 00:00:00+00:00	2024-09-30 00:00:00+00:00	True	500245797.21	no purchase	not applicable
2024-10-09 18:03:49.941000+00:00	2024-10-09 17:05:57.487000+00:00	2024-10-09 18:03:49.941000+00:00	2024-10-07 00:00:00+00:00	False	500245797.22	purchased	recovered
NaT	NaT	2024-10-21 00:00:00+00:00	2024-10-09 18:03:49.941000+00:00	False	500245797.23	no purchase	not applicable
NaT	NaT	2024-10-28 00:00:00+00:00	2024-10-21 00:00:00+00:00	False	500245797.24	no purchase	not applicable
NaT	2024-10-31 19:33:40.426000+00:00	2024-11-04 00:00:00+00:00	2024-10-28 00:00:00+00:00	True	500245797.25	no purchase	not applicable

Cart Abandonment Flag

Process: Append flags onto google_analytics table

ting_system	event_page_name	event_page_title	items	event_date	event_ts_utc	abandoned	purchase_segment	False_by_purchase	recovered
Android	Mycoke Orders - Cart	Cart	[{'item_id': '104631', 'quantity': '1'}, {'ite...	2024-10-08	2024-10-08 17:29:23.925000+00:00	True	5.002458e+08	no purchase	not applicable
Android	Mycoke Orders - Checkout: Review Order	Checkout	[]	2024-10-08	2024-10-08 17:29:34.424000+00:00	True	5.002458e+08	no purchase	not applicable
Android	Mycoke Orders - Checkout: Review Order	Checkout	[]	2024-10-08	2024-10-08 17:29:34.424000+00:00	True	5.002458e+08	no purchase	not applicable
Android	Mycoke Dashboard	Home	[]	2024-10-20	2024-10-20 16:55:07.482000+00:00	False	5.002458e+08	purchased	recovered
Android	Mycoke Orders - Cart	Cart	[]	2024-10-20	2024-10-20 16:55:12.177000+00:00	False	5.002458e+08	purchased	recovered
Android	Mycoke Orders - Cart	Cart	[]	2024-10-20	2024-10-20 16:55:12.177000+00:00	False	5.002458e+08	purchased	recovered
Android	Mycoke Dashboard	Home	[]	2024-10-20	2024-10-20 16:55:12.177000+00:00	False	5.002458e+08	purchased	recovered

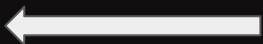
Topics and Findings Q1:

What behavioral events or sequence of events are the strongest predictors of cart abandonment?



Sequence mining process

1. Contextualized generic entries in the **event_name** column with the **event_page_name** column
 - a. Ex. “Page_view” and “button_click” events
2. Filter for segments with the abandoned flag
3. Create bins of events in chronological order for each segment.
4. Mine for the top 15 for each of the following
 - a. Full segments
 - b. Bigrams
 - c. Trigrams
 - d. First activities
 - e. Last activities



Actionable Insights



70% of abandoned carts ended with these events.

1. update_cart ← 25%
2. remove_from_cart
3. view_item_list
4. page_view - Unknown Page
5. page_view - MyCoke Dashboard
6. Button_click - MyCoke Orders - Cart
7. Page_view - MyCoke Orders
8. Button_click - MyCoke Orders - Checkout: Review Order
9. proceed_to_checkout



75% of normal purchases
took place within **1 day** of
these events taking place



Topics and Findings Q2:

What behaviors or conditions lead to a customer returning to complete a previously abandoned cart?

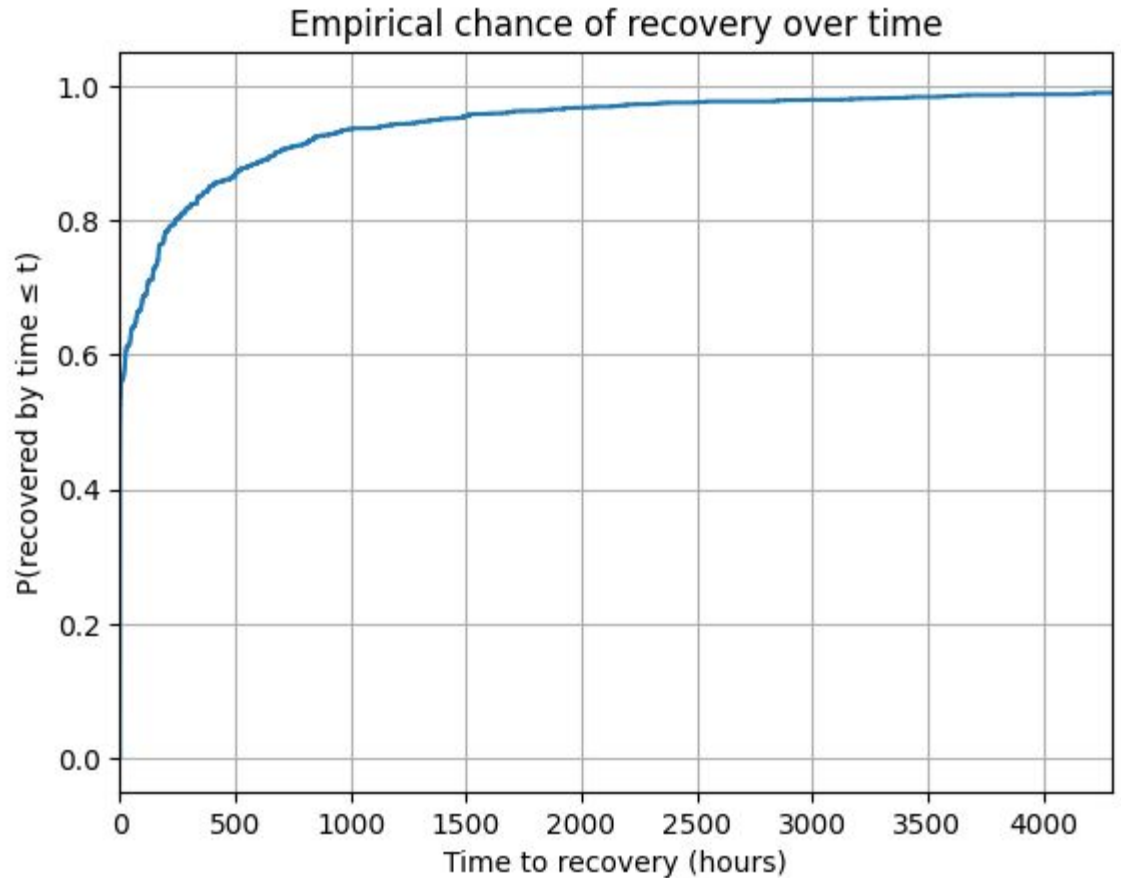


Cart Abandonment is Intentional

Over **50%** of recovered
carts are recovered in 2
hours

+15% (65% total)
recovered within 3 days

Time is a clear indicator

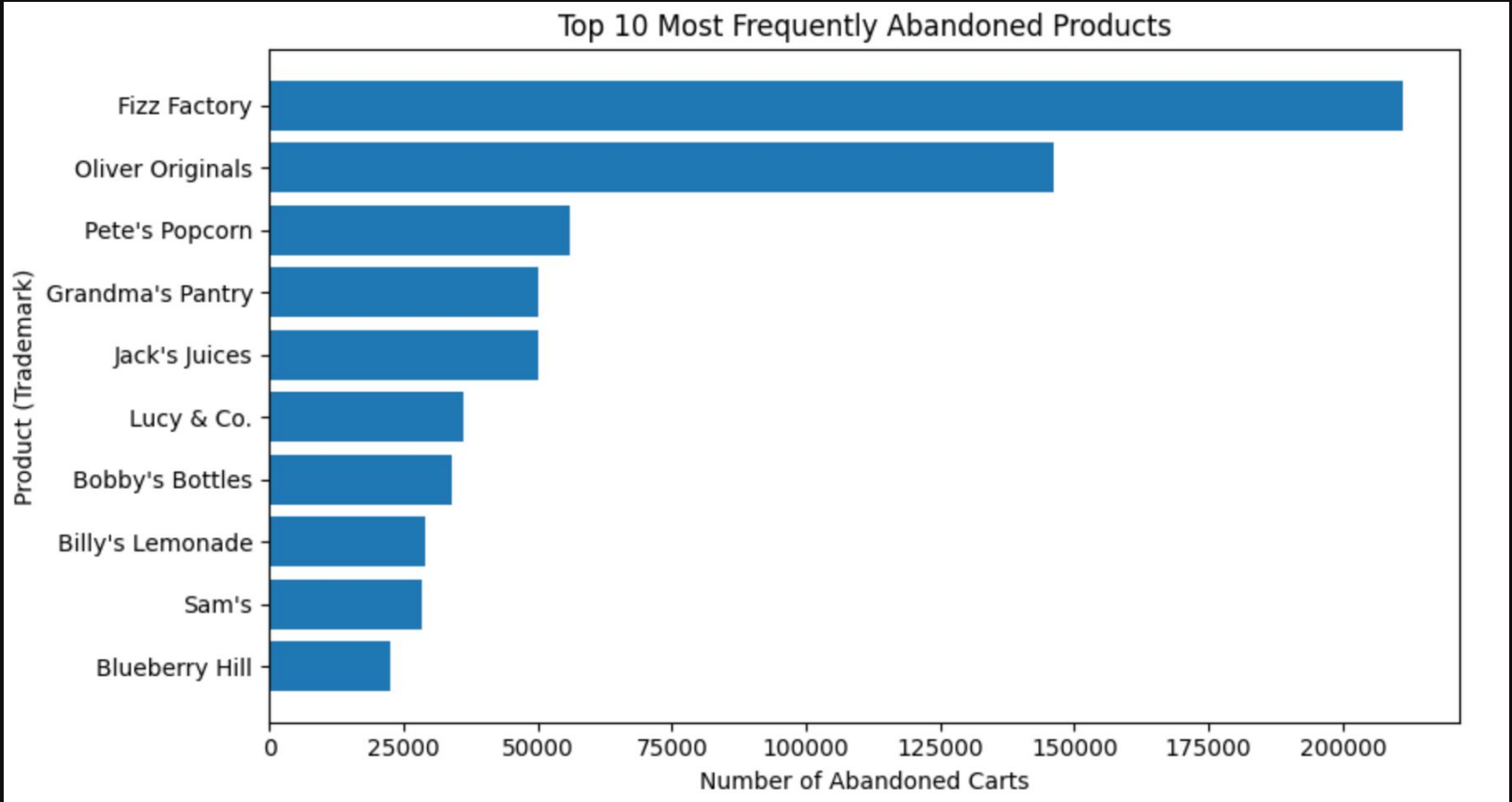


Topics and Findings Q3:

Which products appear most frequently in abandoned carts?



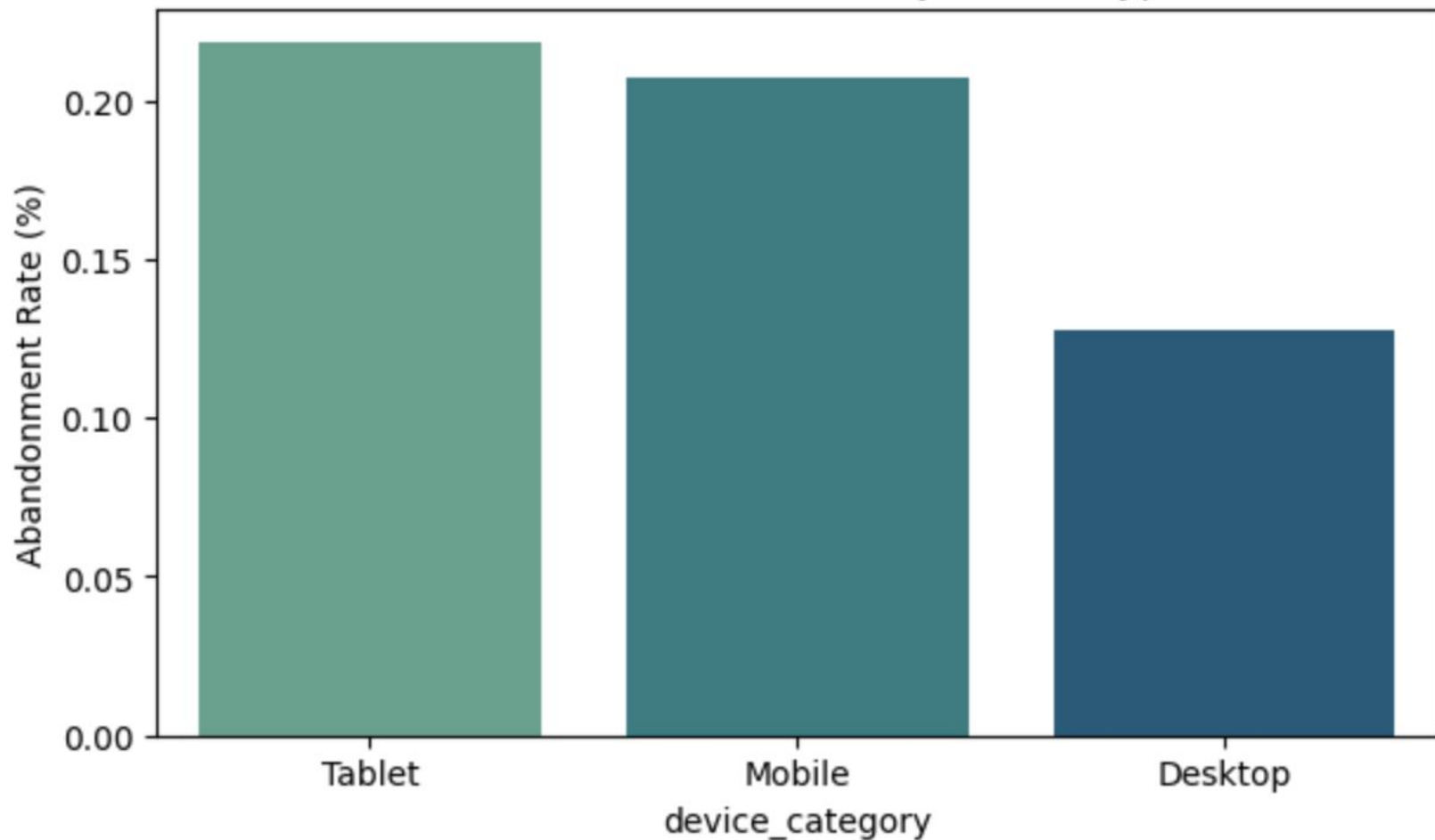
Fizz Factory appears in ~**250,000** abandoned carts



Topics and Findings Q4:
How does cart abandonment vary by
device type and how can it be reduced?



Cart Abandonment Rate by Device Type



Topics and Findings Q5:
What happens after a cart is
abandoned? Does the customer order
through another method or churn?

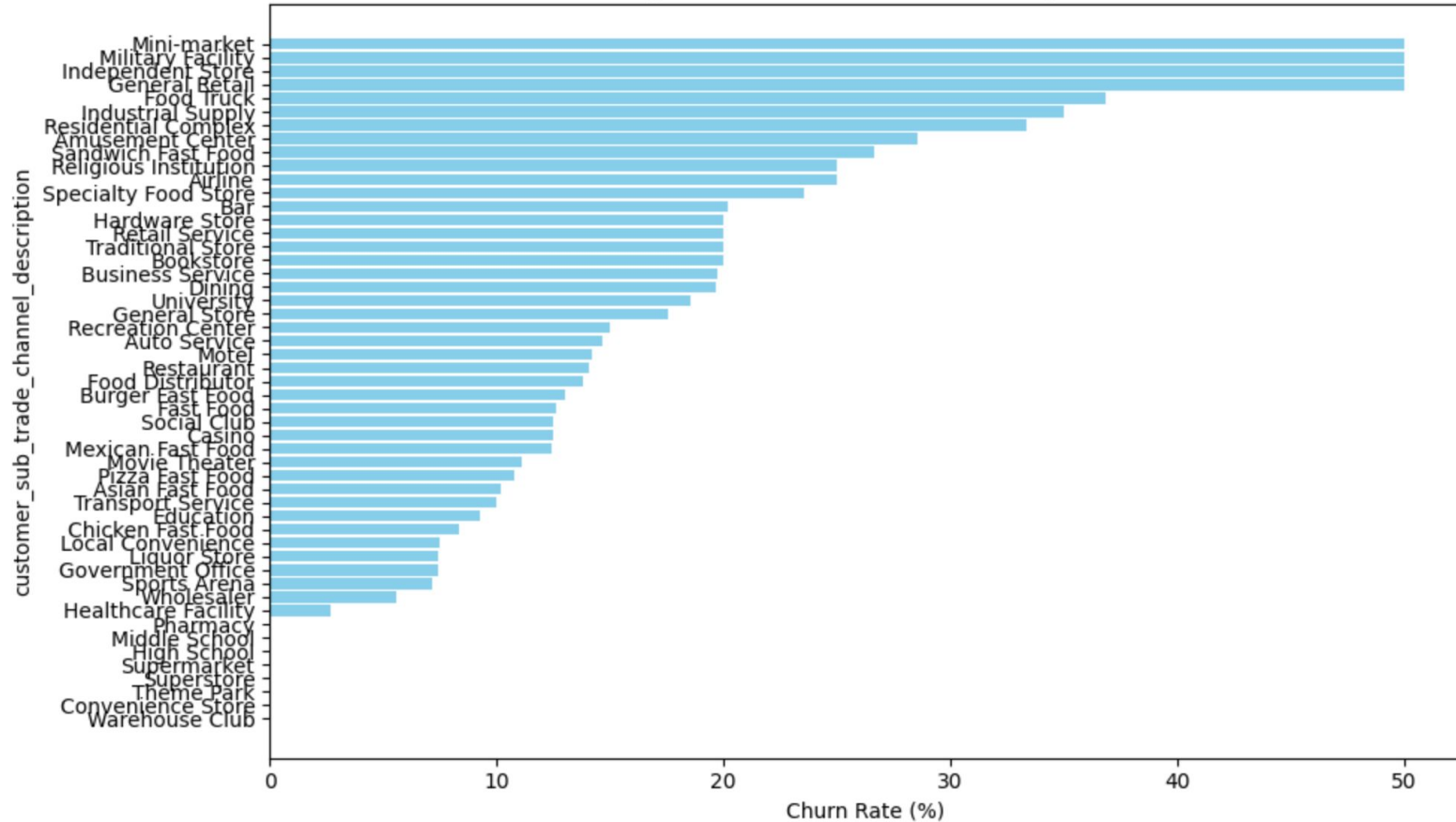


13.5% Churn

- Site interactions, Order volume, and Cart abandonment
- Glenwood Springs, CO; Alamosa, CO; Cheyenne, WY

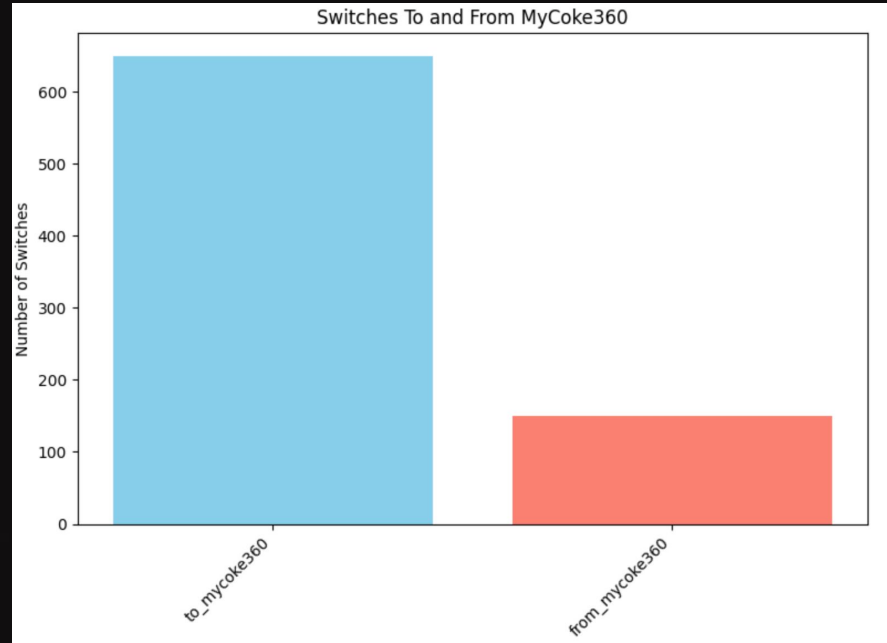


Churn Rate Among Customers Who Abandoned, by Sub Trade Channel



After abandonment:

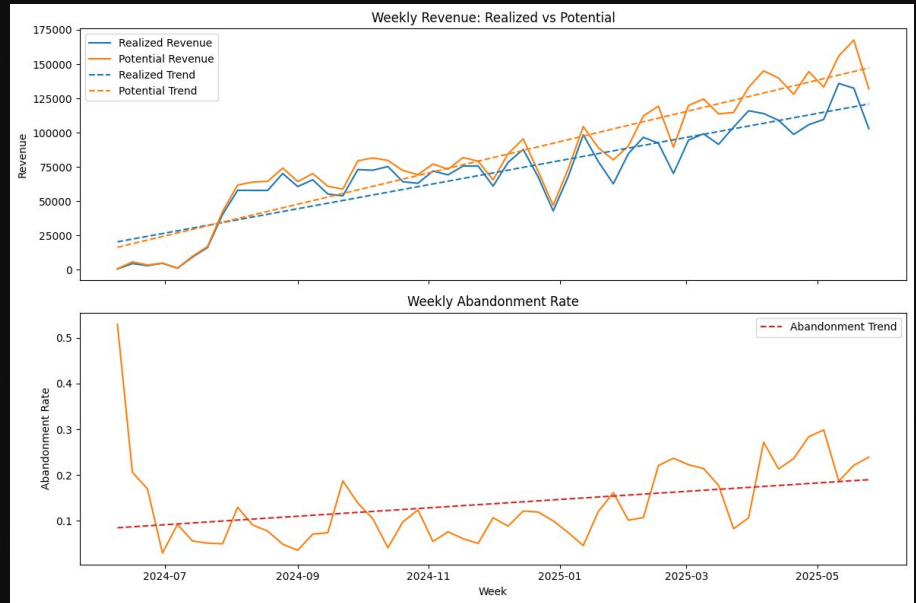
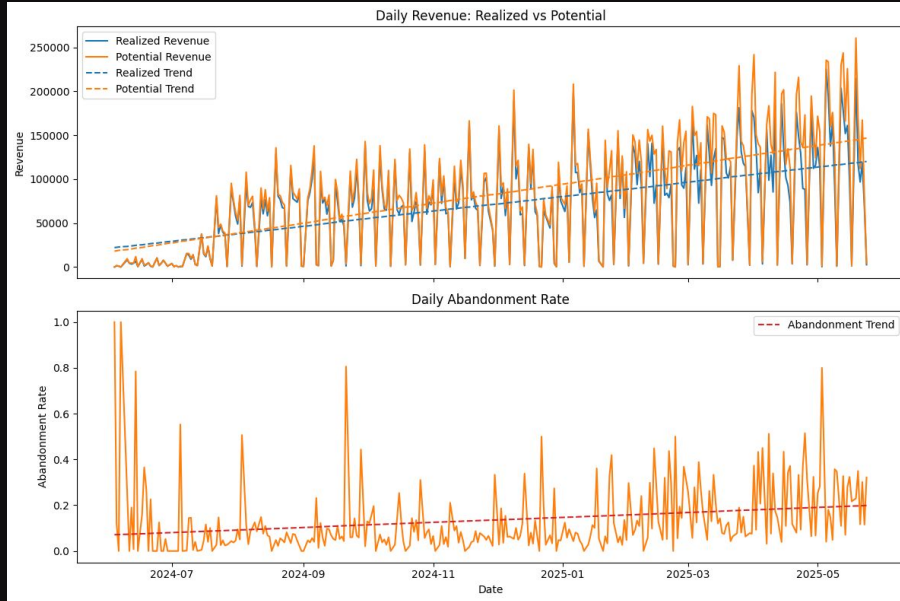
- **649** clients used MyCoke360 as their preferred order method.
- **150** switched.



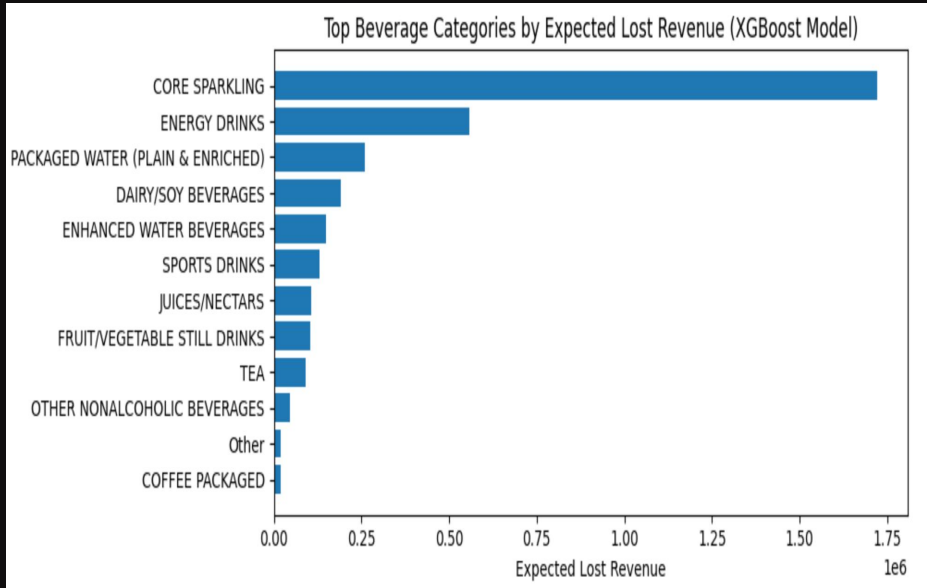
Topics and Findings Q6:
What is the financial impact of cart
abandonment on total MyCoke360
revenue and product mix?



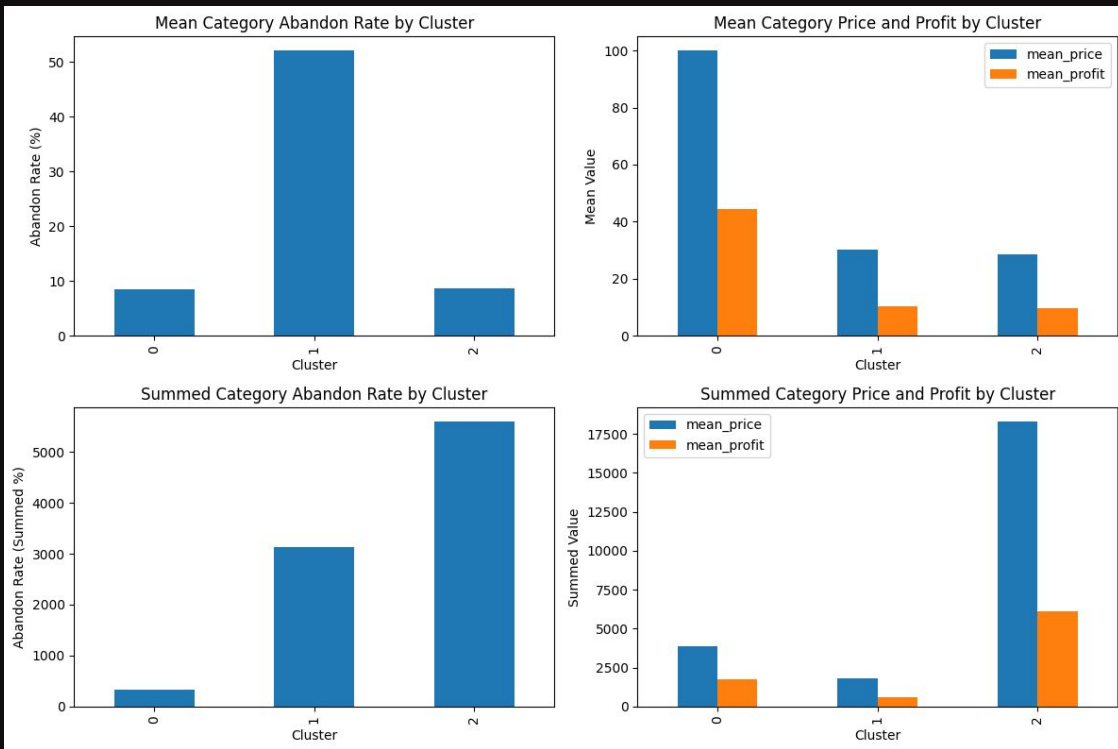
- Total Potential Revenue: **\$21.8** million realized
- Lost profit: **\$1.45** million



Core Sparkling, Energy Drinks, and Packaged Water account
for roughly **70%** of total lost revenue



Category	Mix Shift
CORE SPARKLING	0.004711
ENERGY DRINKS	-0.008688
PACKAGED WATER (PLAIN & ENRICHED)	0.005577
DAIRY/SOY BEVERAGES	-0.001096
SPORTS DRINKS	0.000201
ENHANCED WATER BEVERAGES	-0.002337
FRUIT/VEGETABLE STILL DRINKS	0.001312
JUICES/NECTARS	-0.00095
TEA	0.0004
OTHER NONALCOHOLIC BEVERAGES	0.00066



Cluster 0:

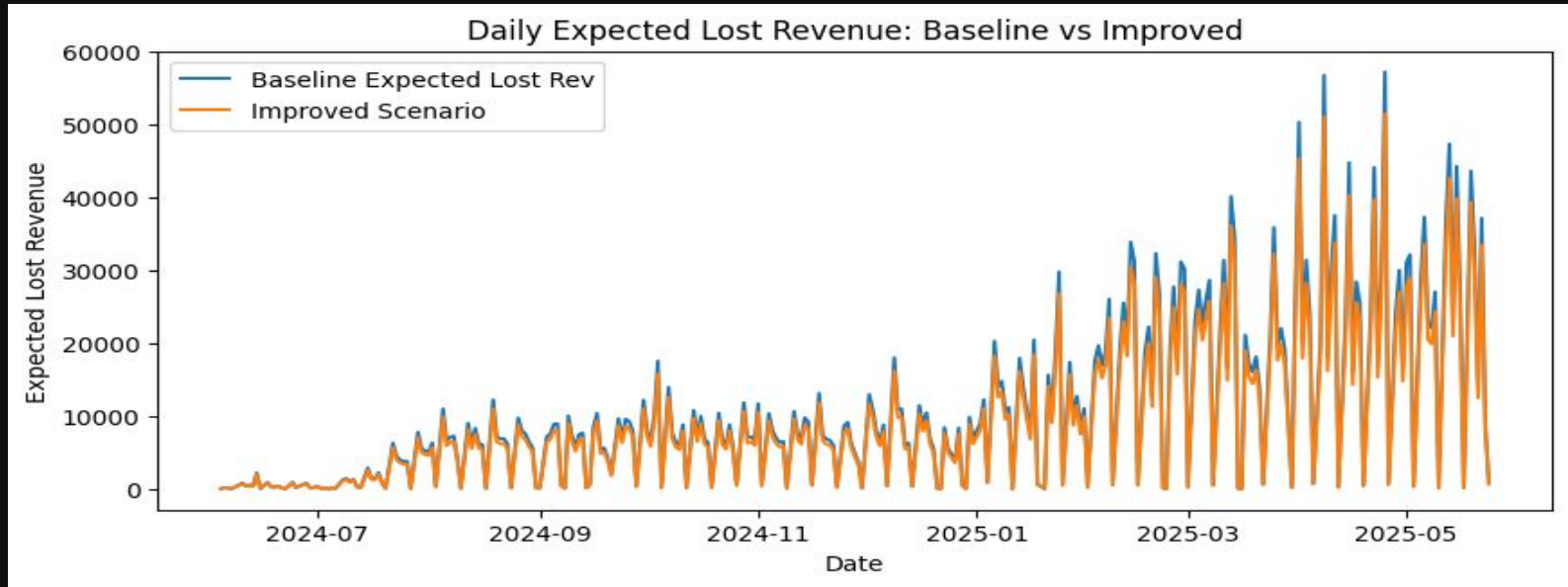
- High category profit items
→ **Each lost sale matters**

Cluster 1:

- Highest category abandonment
→ **Biggest revenue loss**

Cluster 2:

- High volume (scales summed values)
→ **Small improvements scale**



- **10% lower abandonment** → **+\$340K revenue, +\$143K profit**
- Recovered revenue → **strong profit margin (~42%)**
- Improved scenario remains **below baseline** across the full period
- Lower abandonment directly **reduces expected lost revenue**
- **Savings grow** as overall sales volume increases over time

Recommendations



Timely Cart Reminders

Recommendation:

Reach out to customers within **1 day** of the top 9 last events before abandonment, **if there are items in their cart**.

Outcome:

If successful, this could reduce cart abandonment by a **maximum of 70%**.

10% - 70% → \$143k - \$1mil / yr



Further Refinements

- Reduce friction in checkout (especially for mobile and tablet)
- Use predictions to recover at-risk carts
- Tailor strategies by product cluster
- Run A/B tests to validate effectiveness



Q&A





Appendix



Q1



Top FIRST events (abandoned vs completed):

pattern	count_abandoned	percent_abandoned	count_completed	percent_completed
page_view - Mycoke Dashboard	2805.0	70.79757698132256	17933.0	55.408620423296775
session_start	261.0	6.587582029278143	2026.0	6.259848601884752
page_view - Mycoke Orders - Purchase Success	233.0	5.880868248359414	3935.0	12.158195581646842
page_view - Mycoke Orders	131.0	3.3064109035840485	1897.0	5.8612698903136105
page_view - Unknown Page	116.0	2.92781423523473	976.0	3.0156032751429014
nav_link_click	59.0	1.4891468955073195	691.0	2.1350224007415415
button_click - Mycoke Dashboard	54.0	1.3629480060575467	null	null
page_view - Mycoke Order History	47.0	1.1862695608278648	640.0	1.977444770585509
first_visit	43.0	1.0853104492680465	null	null
user engagement	41.0	1.0348308934881372	559.0	1.7271744168082808

Top LAST events (abandoned vs completed):

Top LAST events (abandoned vs completed):

pattern	count_abandoned	percent_abandoned	count_completed	percent_completed
update_cart	1027.0	25.921251892983342	58.0	0.17920593233431176
remove_from_cart	318.0	8.026249369005553	null	null
view_item_list	295.0	7.445734477536598	null	null
page_view - Unknown Page	274.0	6.9156991418475515	null	null
page_view - Mycoke Dashboard	226.0	5.704189803129733	null	null
user_engagement	226.0	5.704189803129733	null	null
button_click - Mycoke Orders - Cart	216.0	5.451792024230187	null	null
page_view - Mycoke Orders	170.0	4.290762241292277	null	null
button_click - Mycoke Orders - Checkout: Review Order	126.0	3.180212014134275	53.0	0.16375714506411246
proceed to checkout	126.0	3.180212014134275	null	null

Top FULL sequences (abandoned vs completed):

[illegible]

Top BIGRAMS (abandoned vs completed):

	pattern	count_abandoned	percent_abandoned	count_completed	percent_completed
	List(view_item_list, view_item_list)	30972.0	7.592590776713309	164818.0	6.22554953927277
	List(page_view - Mycoke Dashboard, page_view - Mycoke Dashboard)	13047.0	3.198389896157127	73127.0	2.762172585266172
	List(page_view - Mycoke Orders, page_view - Mycoke Orders)	11142.0	2.7313911414871397	76082.0	2.8737896349121512
	List(view_item_list, button_click - Mycoke Orders)	10603.0	2.5992586854414057	65954.0	2.491232112470703
	List(button_click - Mycoke Orders, button_click - Mycoke Orders)	10453.0	2.56248712995558	51764.0	1.9552436405666598
	List(page_view - Unknown Page, page_view - Unknown Page)	8616.0	2.1121581471058337	46839.0	1.7692152244900272
	List(button_click - Mycoke Orders, view_item_list)	8472.0	2.0768574538394406	51994.0	1.9639312620281064
	List(page_view - Mycoke Orders, view_item_list)	7889.0	1.9339386748511977	47661.0	1.8002640281478934
	List(view_item_list, add_to_cart)	7340.0	1.7993547817730753	56285.0	2.1260120606849244
	List(add to cart, add to cart)	7123.0	1.7461585981702474	null	null

Top TRIGRAMS (abandoned vs completed):

	pattern	count_abandoned	percent_abandoned	count_completed	percent_completed
	List(view_item_list, view_item_list, view_item_list)	13698.0	3.390913006668944	62943.0	2.4069244535540024
	List(button_click - Mycoke Orders, button_click - Mycoke Orders, button_click - Mycoke Orders)	5993.0	1.4835553839222502	26239.0	1.0033727457668598
	List(view_item_list, view_item_list, button_click - Mycoke Orders)	4871.0	1.2058064867487535	27910.0	1.0672713645471648
	List(page_view - Mycoke Dashboard, page_view - Mycoke Dashboard, page_view - Mycoke Dashboard)	4786.0	1.1847649036295493	28180.0	1.0775960964865319
	List(page_view - Mycoke Orders, view_item_list, view_item_list)	4049.0	1.0023220005842133	22426.0	0.857564586934243
	List(view_item_list, button_click - Mycoke Orders, view_item_list)	3968.0	0.9822706096117951	23369.0	0.8936246692261806
	List(add_to_cart, add_to_cart, add_to_cart)	3644.0	0.9020650457221224	null	null
	List(button_click - Mycoke Orders, view_item_list, button_click - Mycoke Orders)	3451.0	0.8542882746396939	null	null
	List(view_item_list, view_item_list, add_to_cart)	3360.0	0.8317614033003105	23882.0	0.9132416599109778
	List(view item list, button click - Mvcoke Orders, button click - Mvcoke Orders)	3331.0	0.8245825102361113	null	null

Events that on average have a 75% of a purchase taking place within one day of being executed. These are the last events in 70% of abandoned cart segments:

1. update_cart
2. remove_from_cart
3. view_item_list
4. page_view - Unknown Page
5. page_view - Mycoke Dashboard
6. button_click - Mycoke Orders - Cart
7. page_view - Mycoke Orders
8. button_click - Mycoke Orders - Checkout: Review Order
9. proceed_to_checkout

Events that on average have a 85% of a purchase taking place within one day of being executed. These are the last events in 46% of abandoned cart segments:

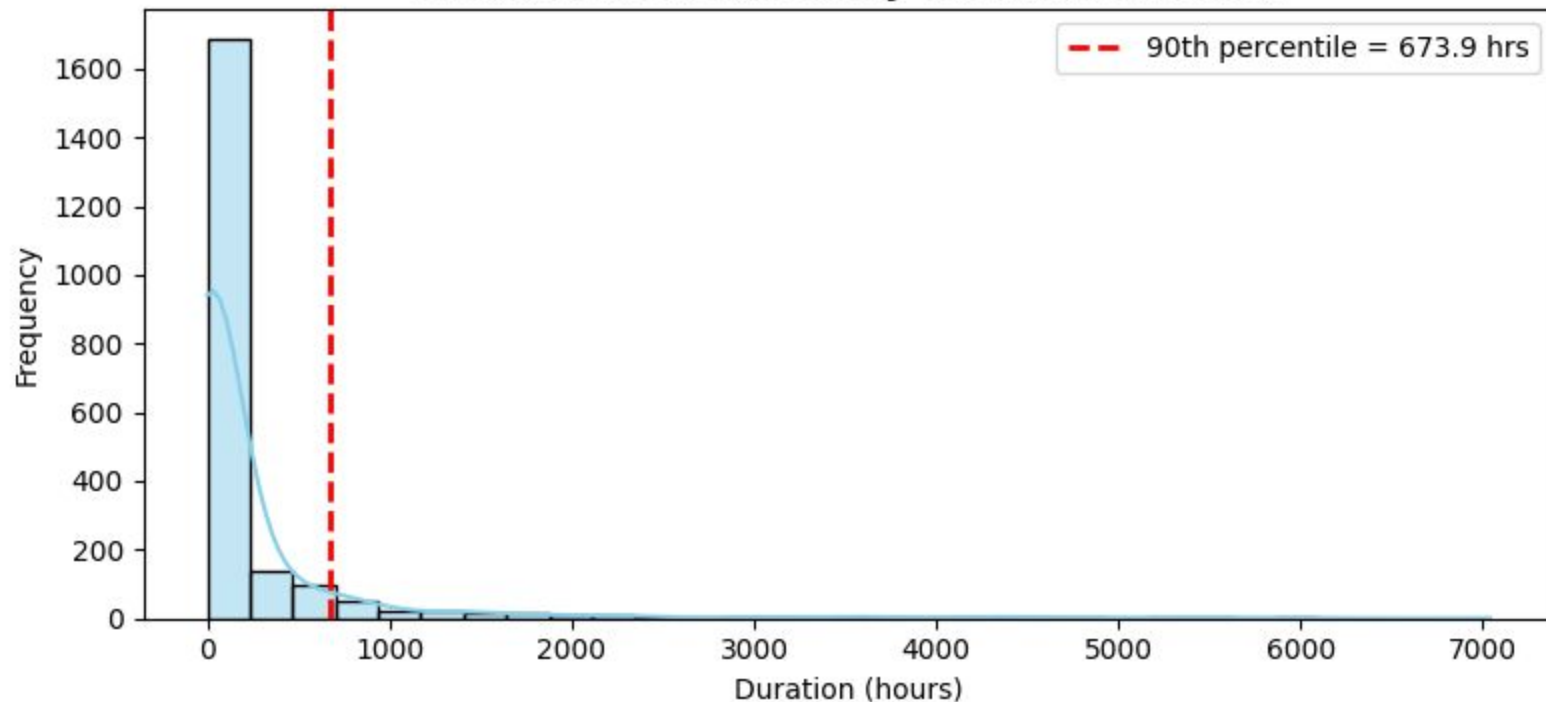
1. update_cart
2. remove_from_cart
3. button_click - Mycoke Orders - Cart
4. button_click - Mycoke Orders - Checkout: Review Order
5. proceed_to_checkout

	pattern	percent_abandoned	75%	85%
	update_cart	25.921251892983342	0.0	0.0
	remove_from_cart	8.026249369005553	0.0	5.0
	view_item_list	7.445734477536598	7.0	36.0
	page_view - Unknown Page	6.9156991418475515	15.0	61.0
	page_view - Mycoke Dashboard	5.704189803129733	18.0	60.0
	user_engagement	5.704189803129733	49.0	84.0
	button_click - Mycoke Orders - Cart	5.451792024230187	0.0	12.0
	page_view - Mycoke Orders	4.290762241292277	23.0	68.0
	button_click - Mycoke Orders - Checkout: Review Order	3.180212014134275	0.0	0.0
	proceed to checkout	3.180212014134275	0.0	0.0

Q2



Distribution of Recovery Durations (Hours)

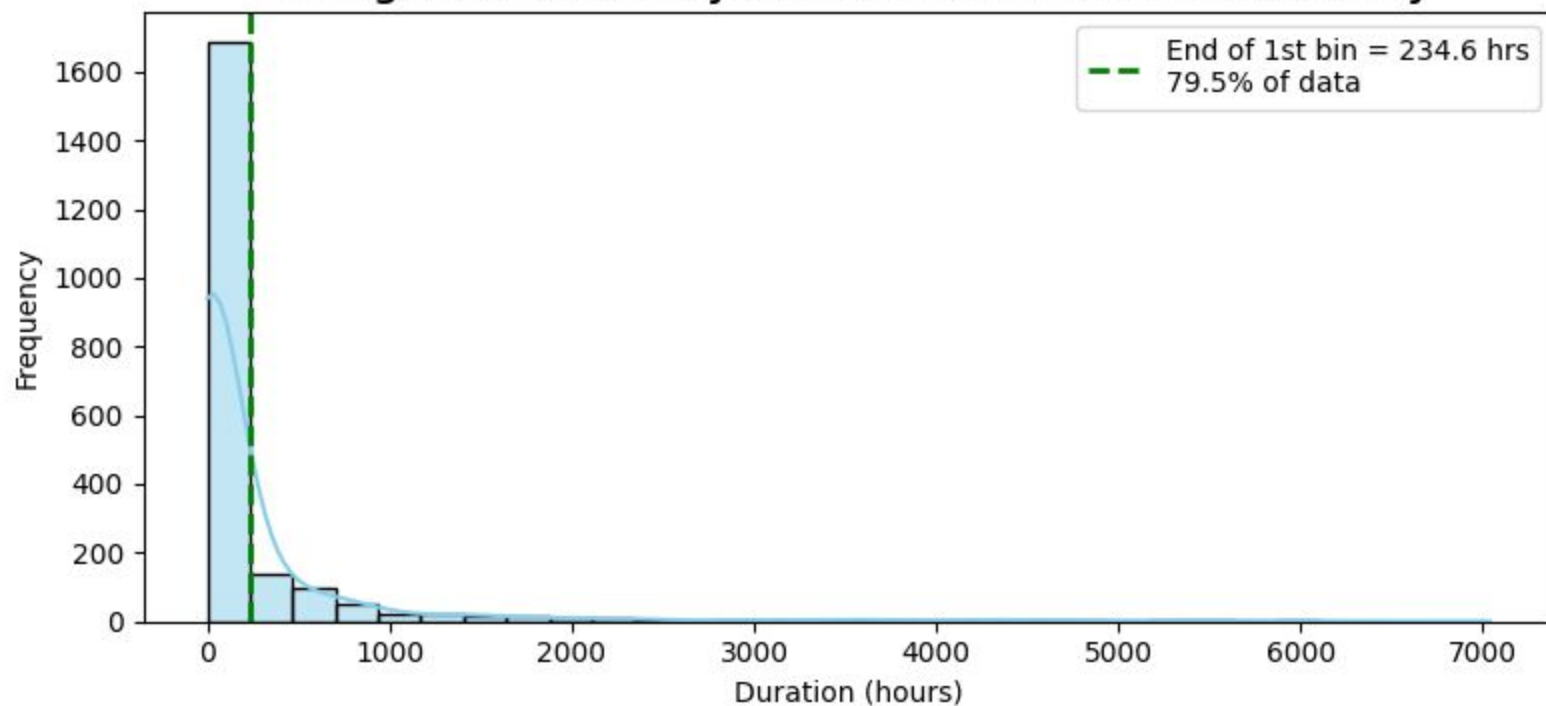


Total observations: 2,123

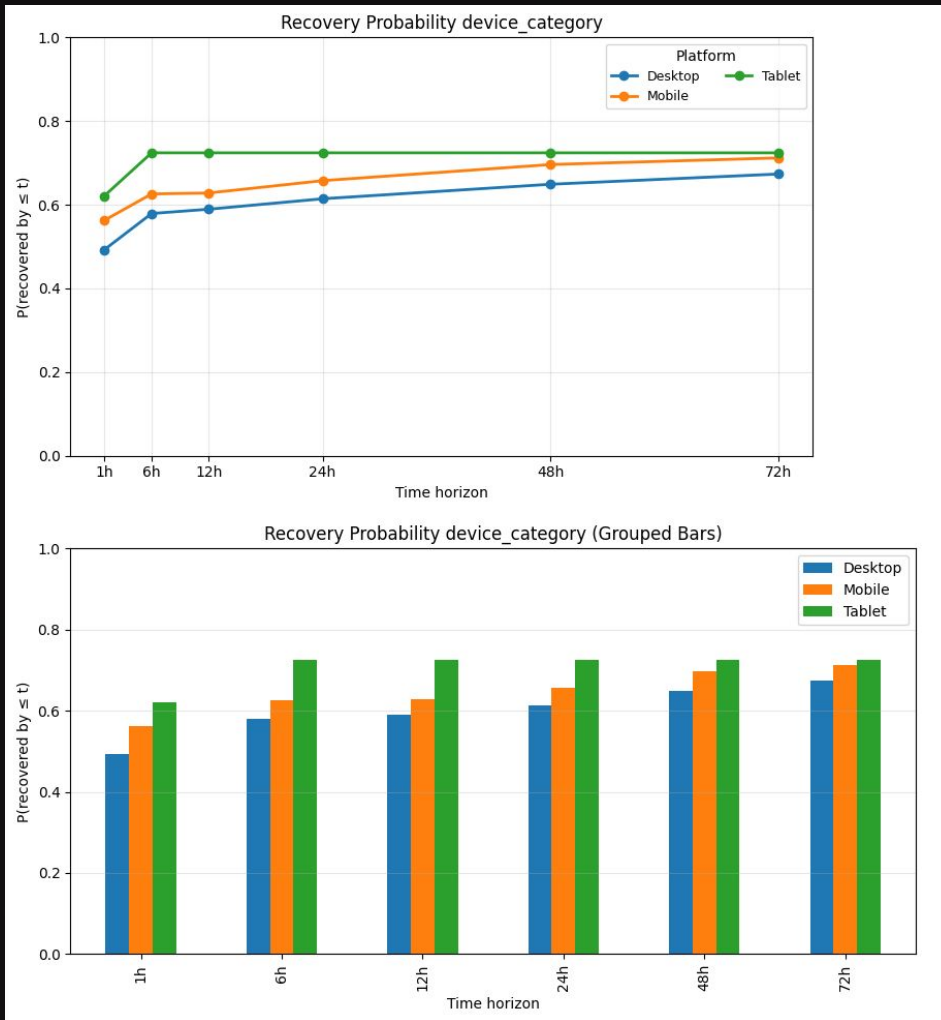
Number of observations in first bin: 1,687 (79.46% of total)

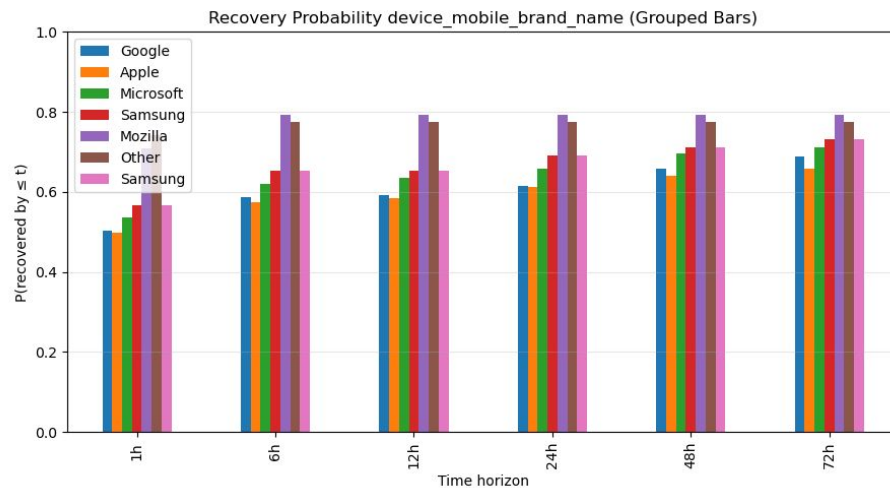
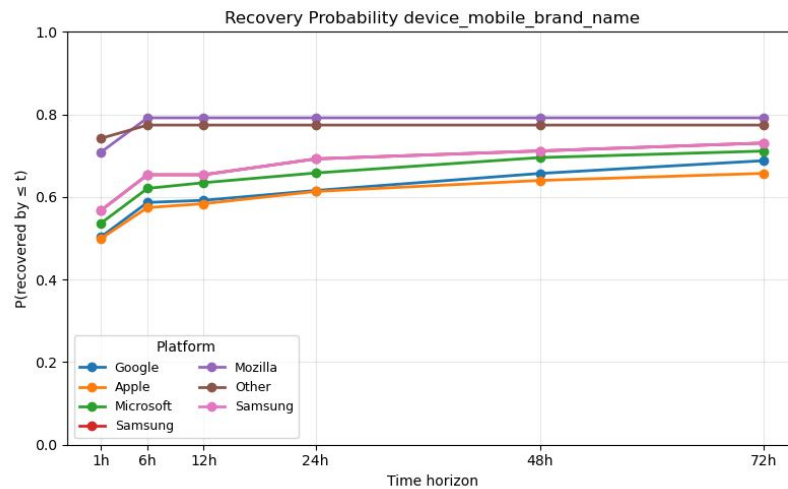
Duration range of first bin: 0.00 to 234.61 hours

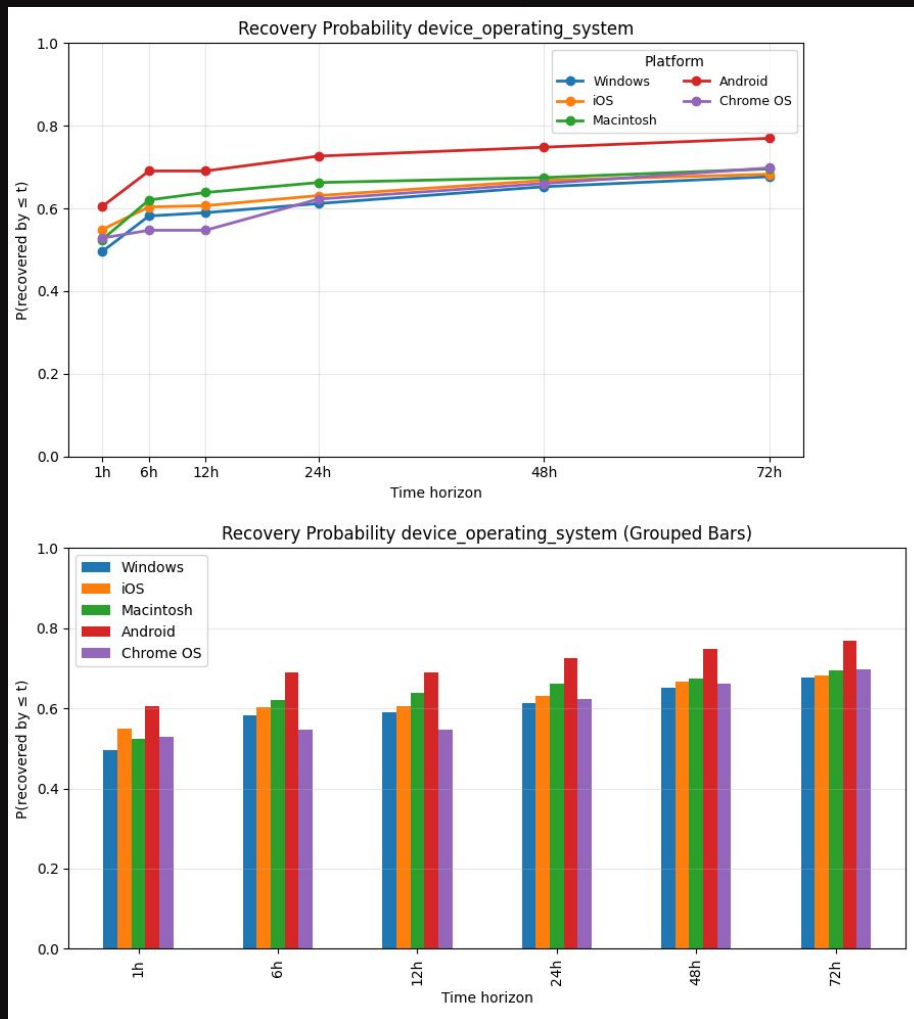
Histogram of Recovery Durations with First Bin Boundary



	P(recovered by $\leq t$)
1h	0.470561
6h	0.554404
12h	0.563354
24h	0.592558
48h	0.629298
72h	0.653792







Q3



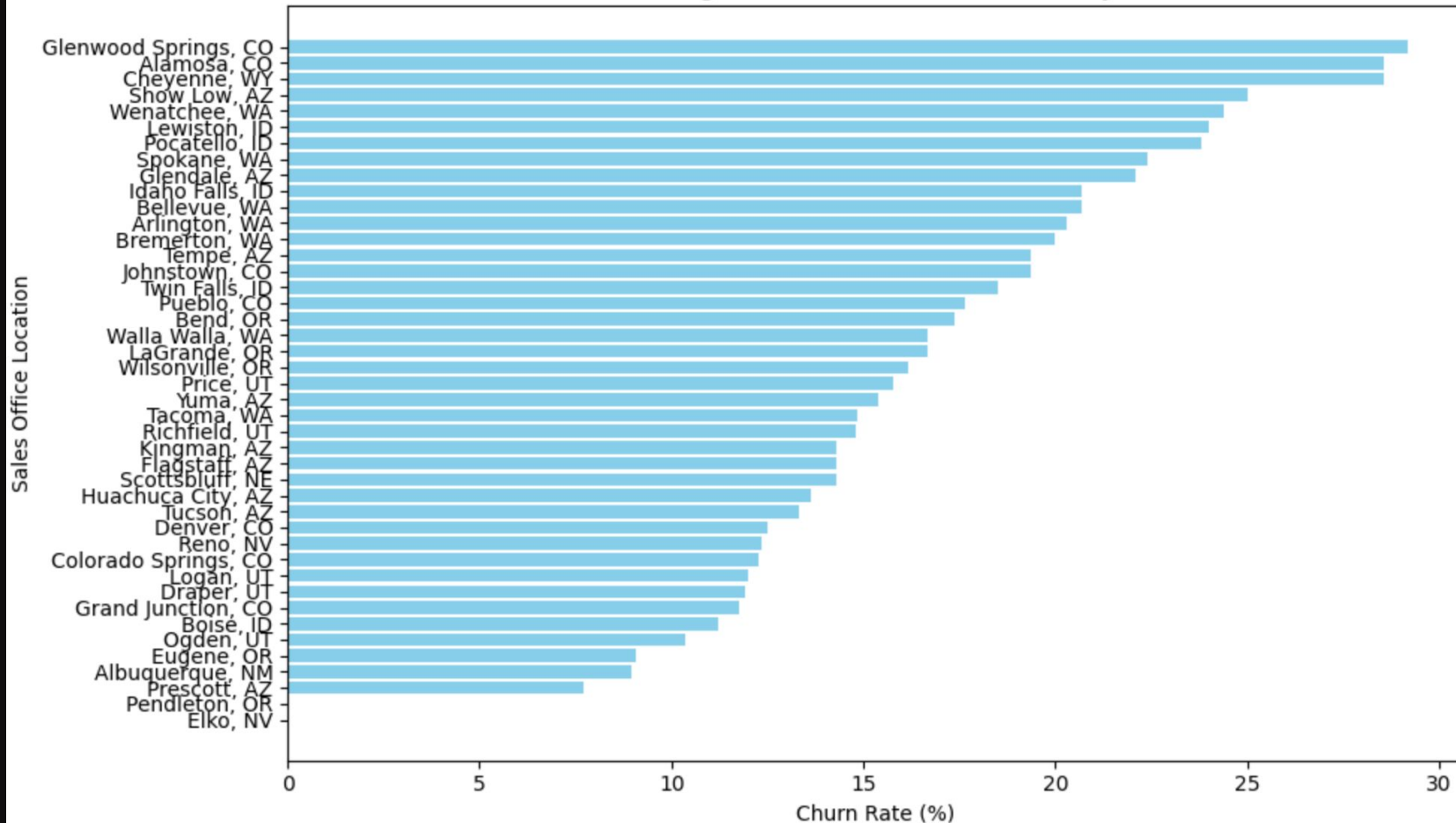
Q4

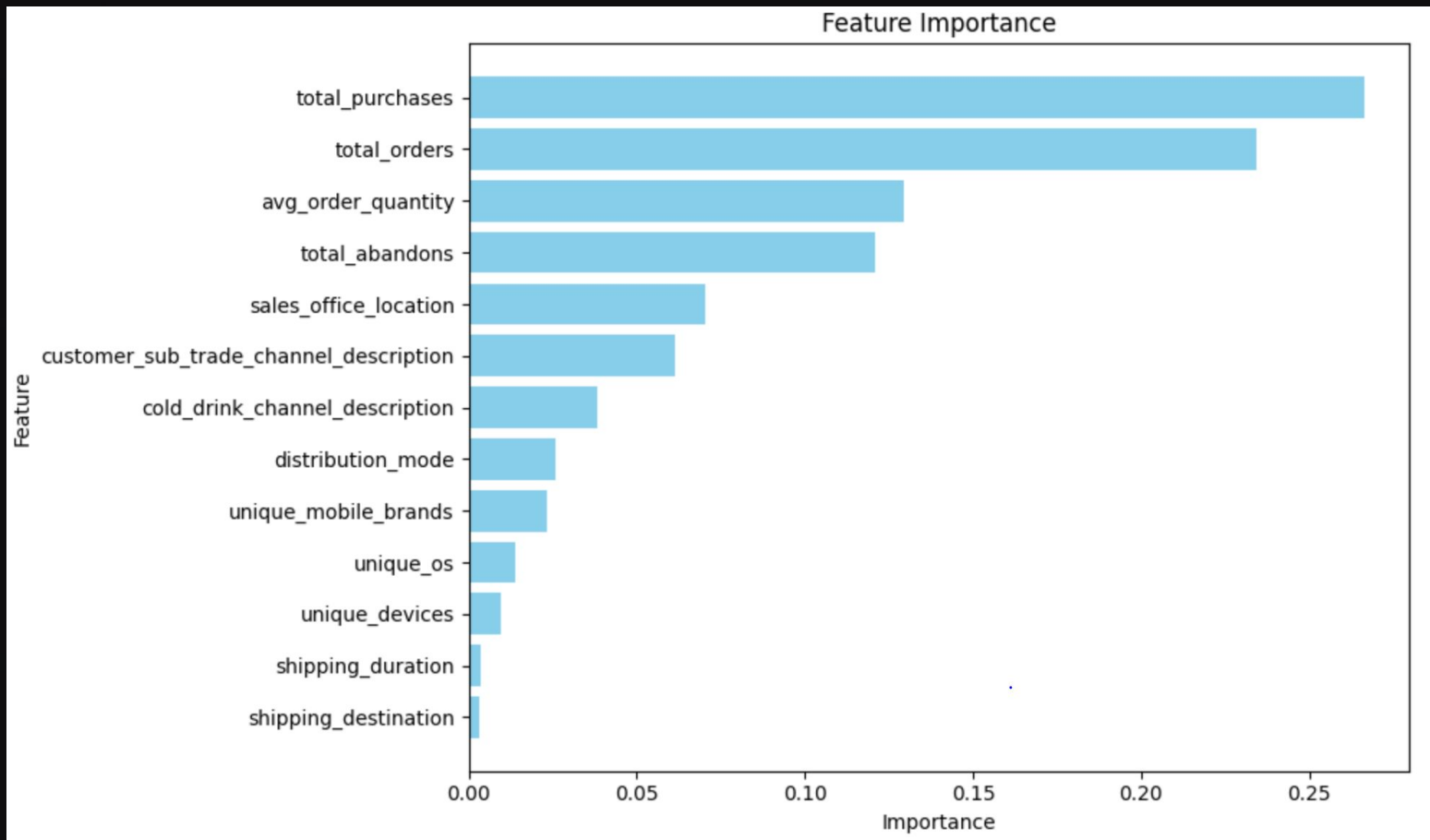


Q5

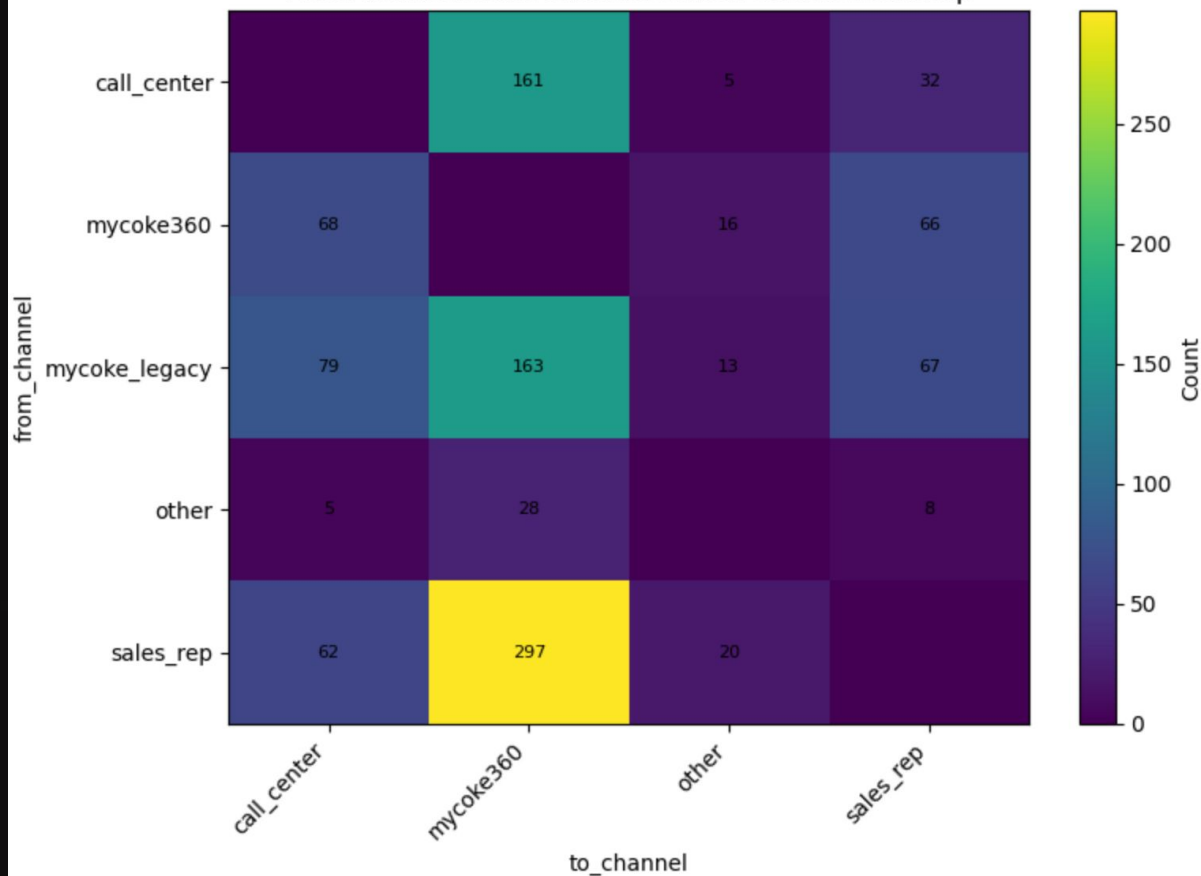


Churn Rate Among Customers Who Abandoned, by Sales Office





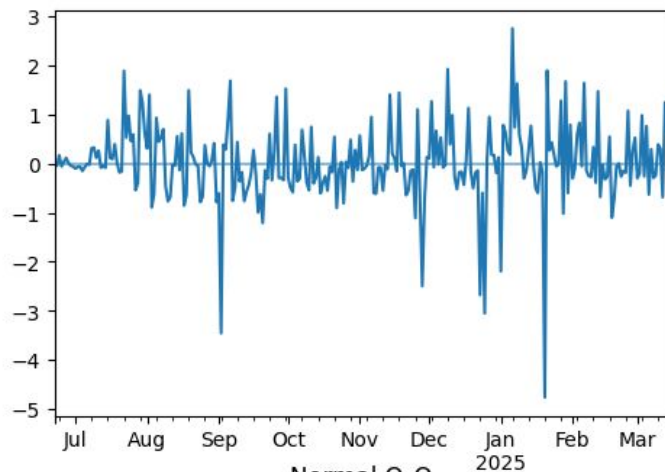
Post Abandonment Channel Switch Counts Heatmap



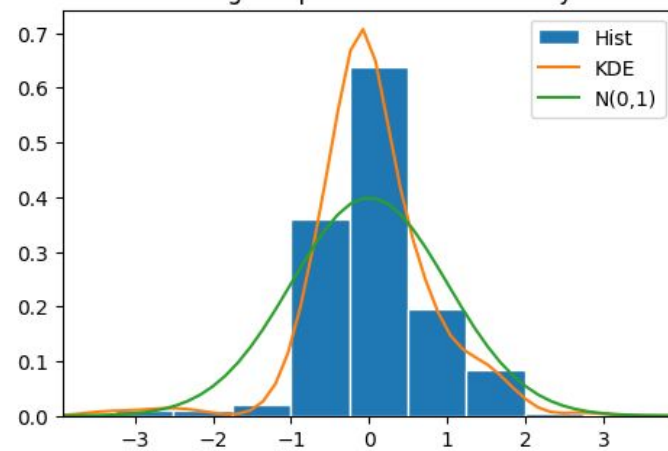
Q6



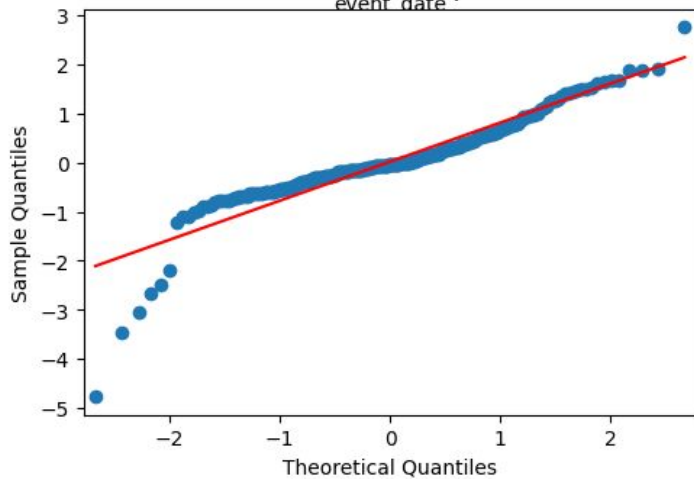
Standardized residual for "r"



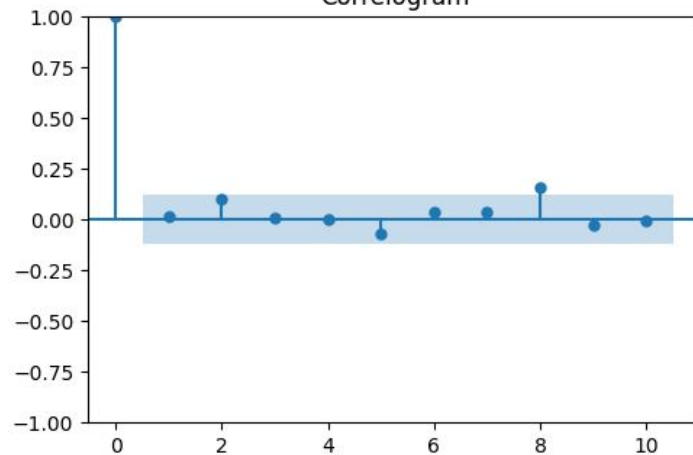
Histogram plus estimated density

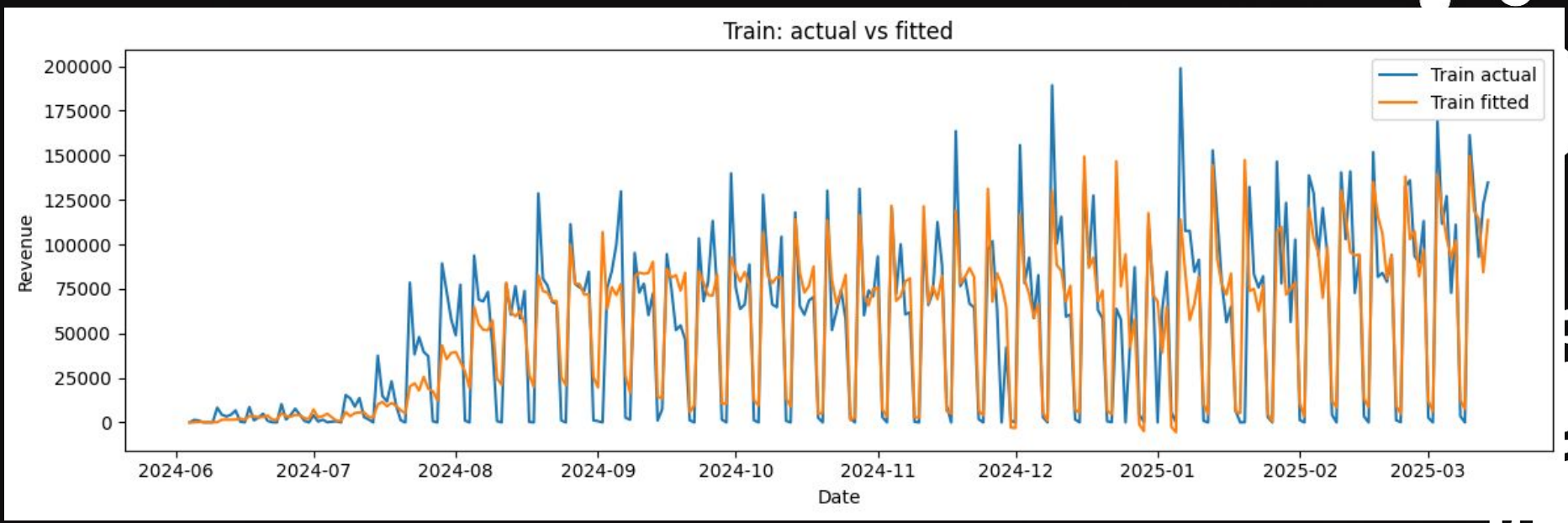


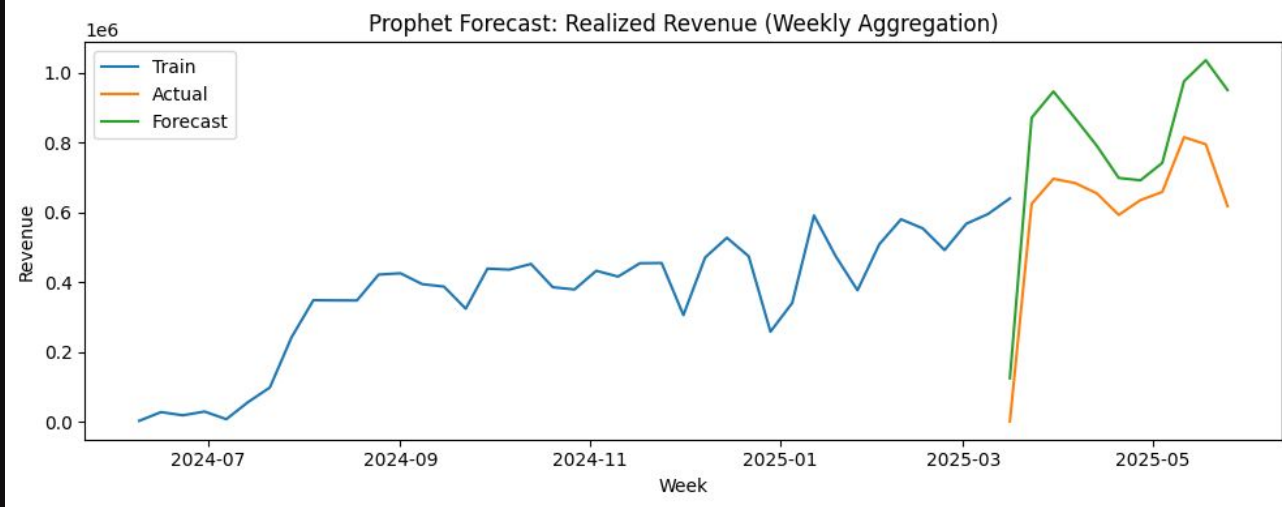
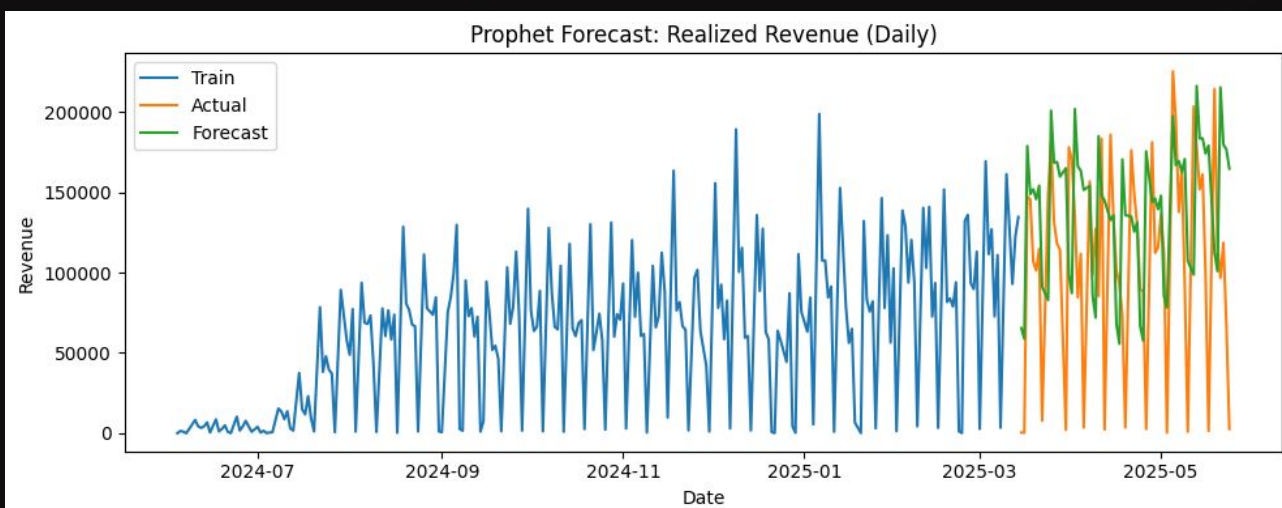
Normal Q-Q



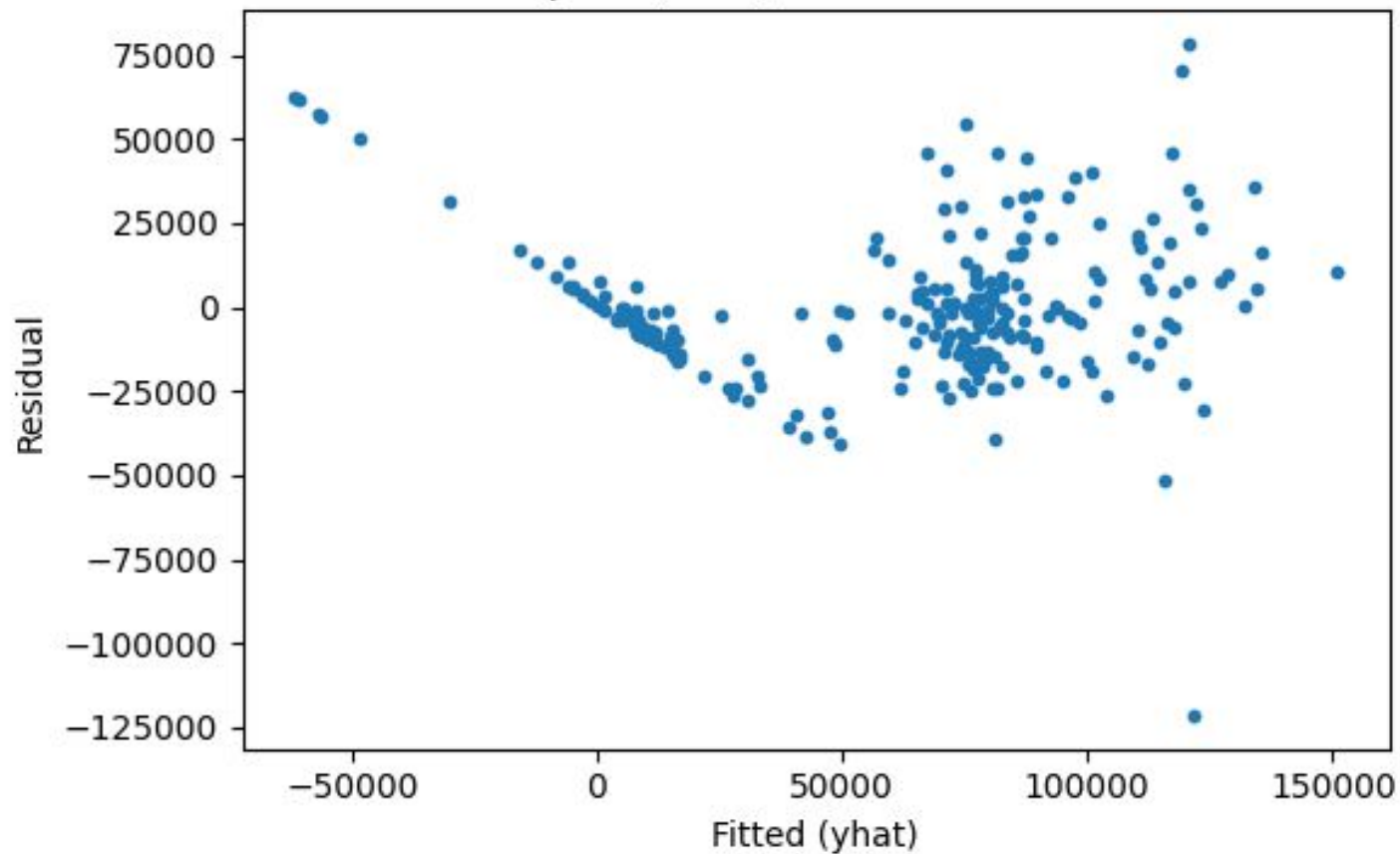
Correlogram

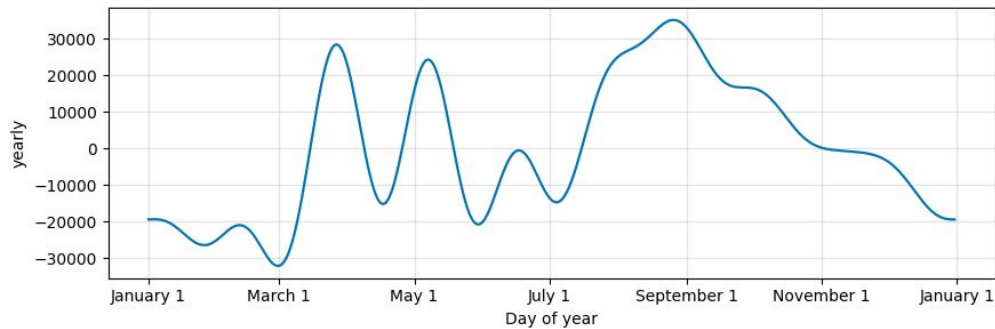
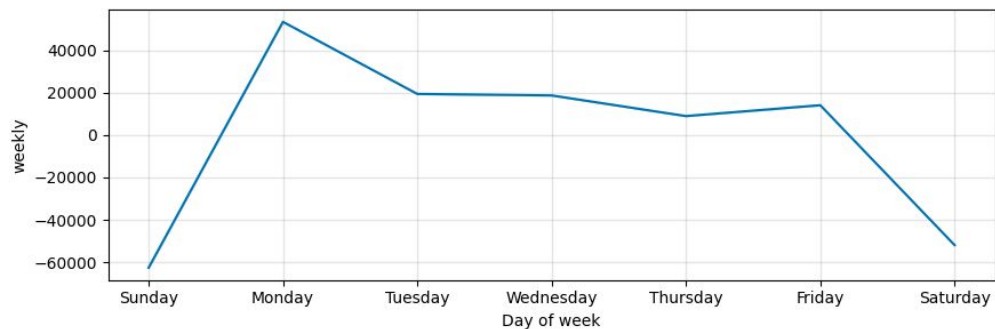
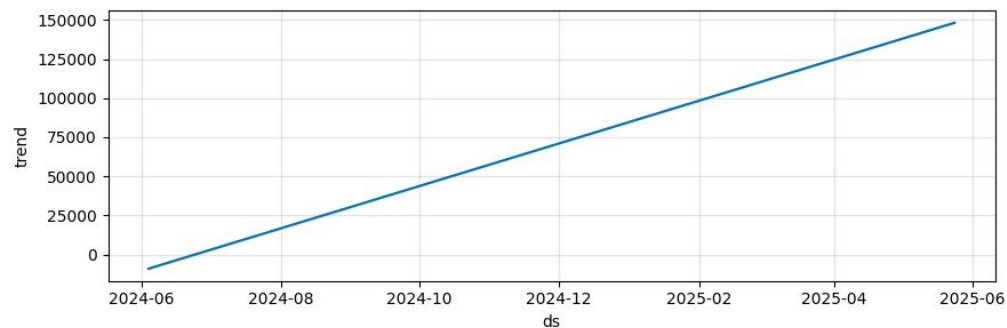


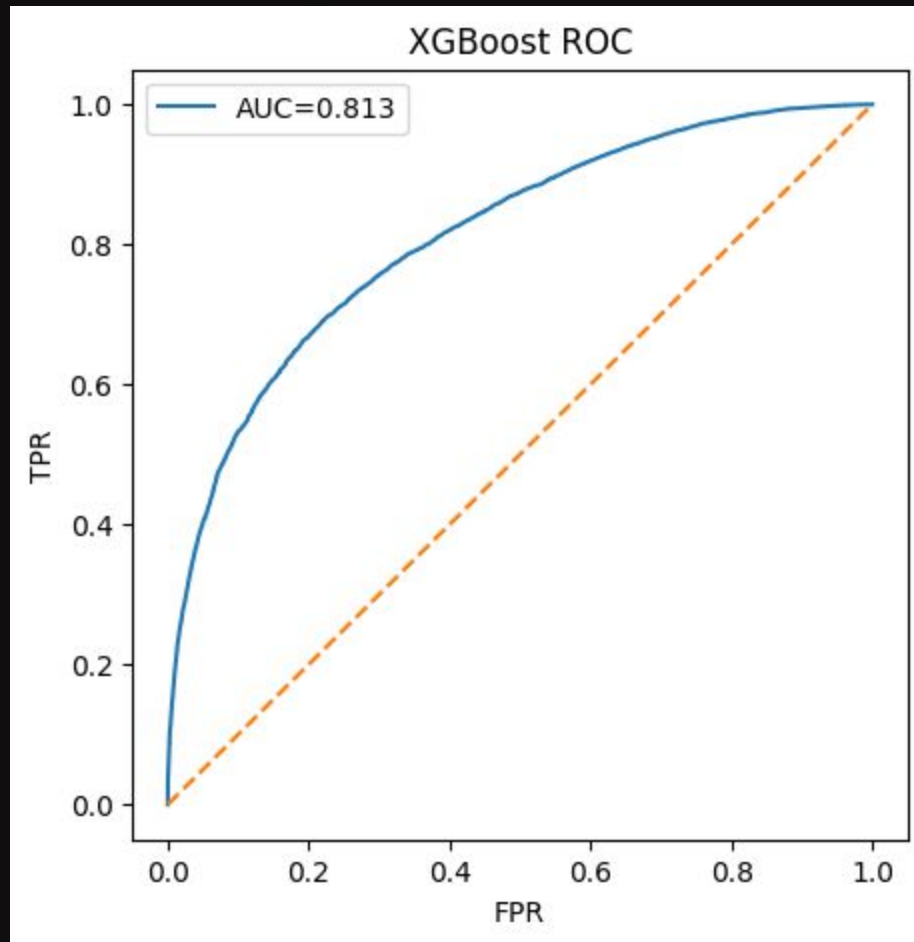


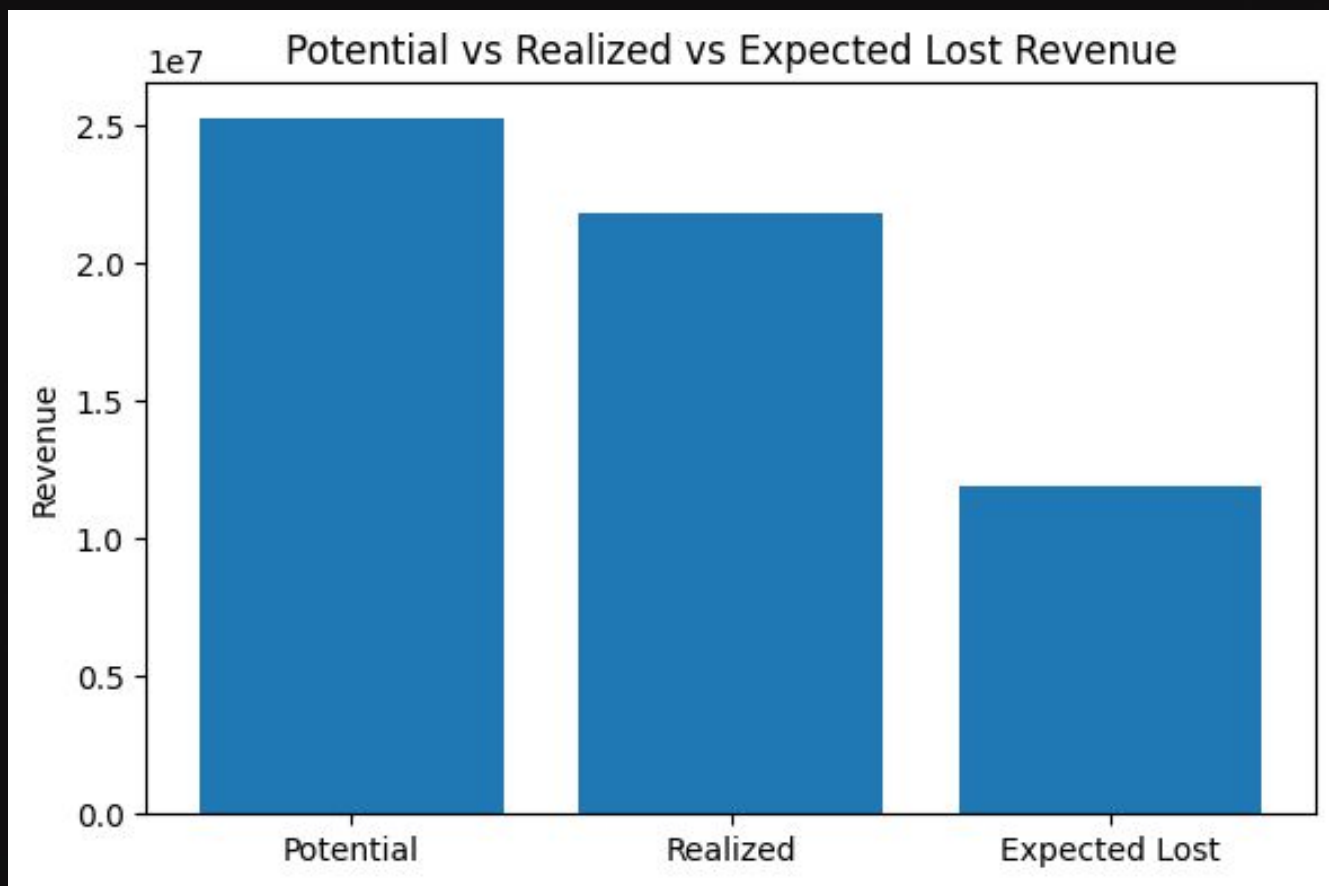


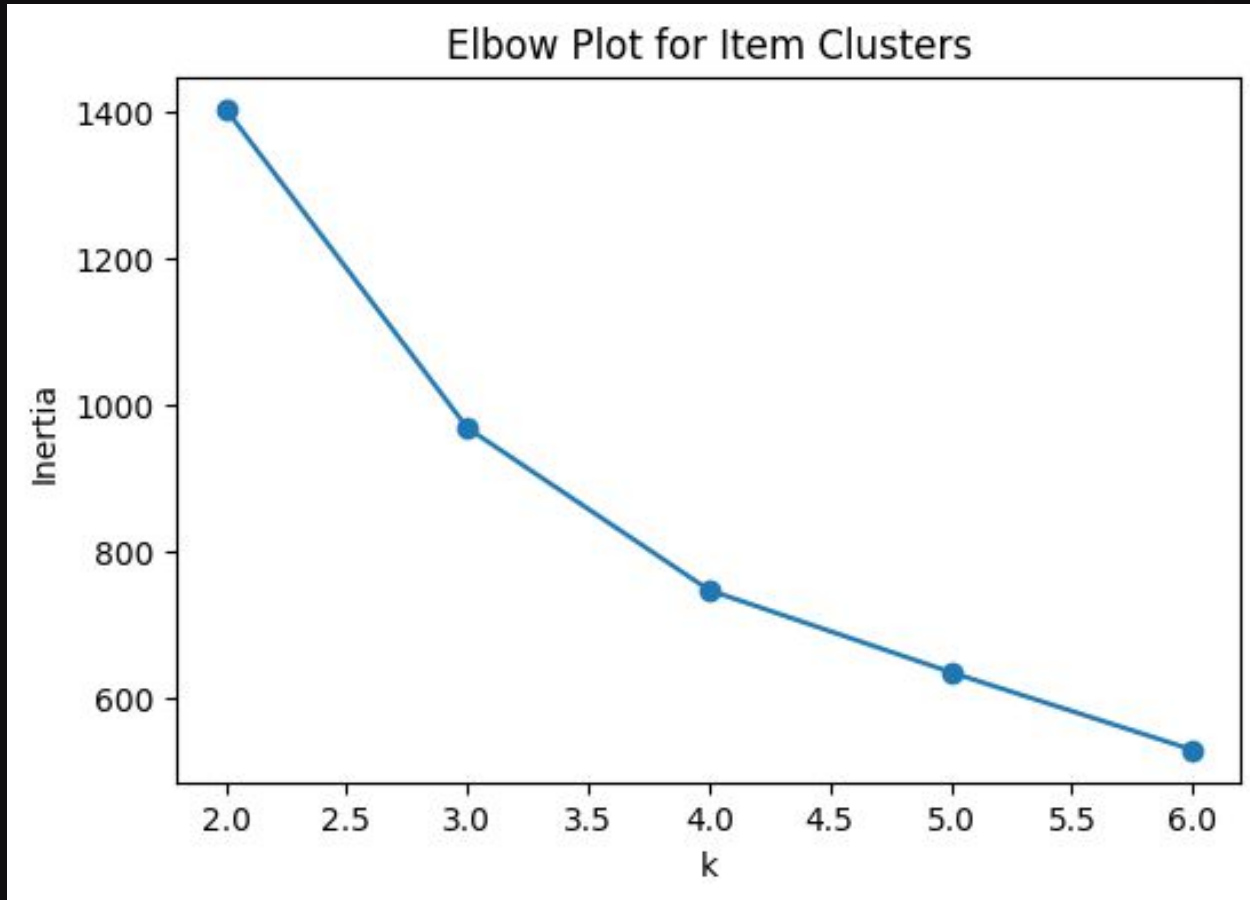
Prophet (train) residuals vs fitted











cluster	category	n_items	avg_abandon_proportion	avg_profit
0	Other	19	0.0832939114462945	50.6358576950764
0	CORE SPARKLING	9	0.0818629695732594	26.464893143101342
0	OTHER NONALCOHOLIC BEVERAGES	4	0.0872873609135985	39.15545690250863
0	FRUIT/VEGETABLE STILL DRINKS	3	0.0896344016284716	60.968860929033575
0	JUICES/NECTARS	3	0.1164710081401279	48.9317907720775
0	PACKAGED WATER (PLAIN & ENRICHED)	1	0.0289855072463768	42.23159420289855
1	ENERGY DRINKS	24	0.6070325488891006	7.290765515203411
1	CORE SPARKLING	15	0.4772881668099868	14.049034198851269
1	SPORTS DRINKS	7	0.4568080728795015	8.190201940916227
1	ENHANCED WATER BEVERAGES	4	0.3878583109023568	9.601365716270717
1	TEA	4	0.5764135190502673	7.896788353062684
1	FRUIT/VEGETABLE STILL DRINKS	2	0.3666666666666667	18.662416666666665
1	PACKAGED WATER (PLAIN & ENRICHED)	2	0.4456140350877193	12.83496052631579
1	DAIRY/SOY BEVERAGES	1	0.3597944031981724	6.000970873786408
1	Other	1	0.4583333333333333	34.190666666666665
2	CORE SPARKLING	258	0.0813861889931345	11.05758144251579
2	ENERGY DRINKS	156	0.0885311707462916	6.693162121945891
2	SPORTS DRINKS	73	0.1027992879799923	8.678843172774345
2	PACKAGED WATER (PLAIN & ENRICHED)	38	0.0728191266326337	10.734249617592234
2	ENHANCED WATER BEVERAGES	36	0.0895012579023681	10.158260482097084
2	FRUIT/VEGETABLE STILL DRINKS	21	0.0984875938641915	11.619954520494437
2	TEA	21	0.0939048686192969	13.213014806500745
2	OTHER NONALCOHOLIC BEVERAGES	14	0.0716348100341365	4.313255702026188
2	DAIRY/SOY BEVERAGES	8	0.1382964367374496	8.974038381929464
2	COFFEE PACKAGED	6	0.0655648552069258	7.434943500428864
2	JUICES/NECTARS	6	0.1581378314834398	12.591947076339414
2	Other	3	0.0574712643678161	18.192816091954025